The Role of Genetics in Fibropapillomatosis Susceptibility in Sea Turtles

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How do genetic, environmental, and behavioral factors interact to influence fibropapillomatosis in green sea turtles, and how can this understanding support more effective conservation and rehabilitation strategies?

Background

- My community
- Green Sea Turtle Endangerment
- What is Fibropapillomatosis (FP)?
- Why does this matter?

Focuses of Study

Genetics

Environmental

Behavior

Genetic Methods

RNA-seq & Whole Genome Sequencing (WGS)

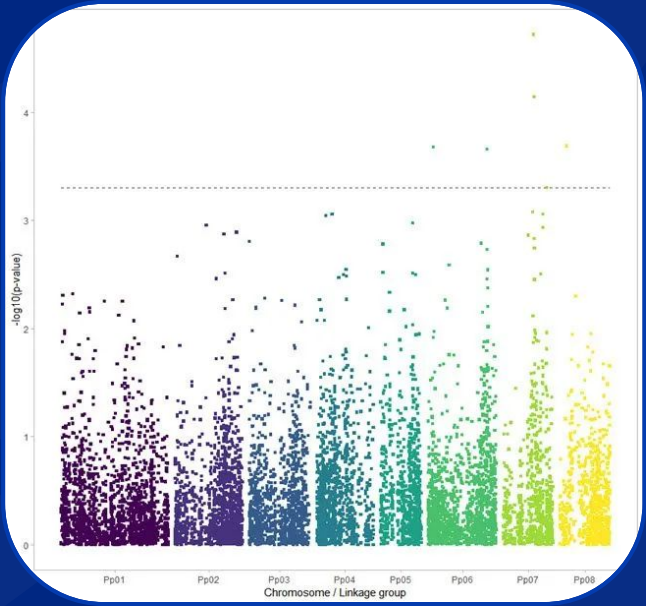
- Compare gene expression in healthy vs. affected turtles

GWAS using SNPs

- Identify genetic variants linked to FP susceptibility

Bioinformatics tools

- Analyze pathways, relate to cancer research (Duffy et al.)



Environmental Methods

Habitat mapping with GIS + satellite data

Water quality testing
- Pollutants, nutrients, temperature

Field surveys with conservation groups
- monitor FP prevalence + real-time data collection

Behavioral Methods

Animal tracking to assess:

- Home rang
- Habitat use
- Foraging and navigation

Compare healthy vs. FP-affected turtles

Data Analysis

Use regression models and data analytics to find correlations

Build predictive models for FP susceptibility

Goal: Identify high-risk populations and plan for targeted conservation efforts

Conservation and Rehabilitation

Identifying High-Risk Populations
- Use genetic + environmental data
- Focus on vulnerable hotspots

Improving Rehabilitation Outcomes
- Prioritize treatable cases (e.g., no eye tumors)
- Tailor care using immune markers

Monitoring & Release Strategies
- Track movement post-rehab
- Adjust release based on behavior data

Guiding Policy & Collaboration
- Share findings with orgs like NOAA
- Support science-based marine policy

Sources

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Thank you!!

Any questions?